

Engineering Markov Blankets: When Sensory States Descend from Active States

Luca M. Possati
University of Twente
l.m.possati@utwente.nl

Draft v9

Abstract

Markov blankets are usually analysed as if the observation channel were exogenous. In that setting, comparing boundary designs reduces to Blackwell’s comparison of experiments. A Markov blanket, however, contains active states, and these can causally influence later sensory states. This paper studies what changes when they do, and turns the answer into a design problem.

In a minimal multi-agent system, one observer echoes the decision agent’s previous announcement with probability κ . We compare this closed loop with an open-loop system built to be *the same experiment at every step*. We prove four results. (i) No criterion defined on per-step channels distinguishes the two systems. (ii) A factorised generative model is exact in one and misspecified in the other. (iii) The closed loop is a sequential garbling of its step-equivalent open loop, so even a perfectly specified agent loses decision value. (iv) An agent that ignores the loop turns its wrong beliefs into attractors above an analytic threshold κ^* . At $\rho = 0.6$, $\kappa = 0.5$, $T = 30$, final accuracy is 98.5% in the open loop, 90.2% in the closed loop for a correctly specified agent, and 76.2% for an agent that ignores the loop. The last ends almost certain of the wrong state in 19.3% of episodes.

We then treat the blanket’s topology as the design variable. A cost-based design map shows when an engineer should leave the loop, cut it, or have the agent learn it. Finally, we let the agent engineer its own blanket: before each episode it may withhold its announcements, cutting the loop at a cost. Cutting the loop also removes all evidence about the loop, so a greedy agent can seal its boundary on the basis of beliefs it can never revise. We prove this lock-in is absorbing. An agent that adds the expected information gain about its own blanket topology to its policy value (an expected-free-energy term) avoids it. At an intermediate cost, this agent reduces cumulative regret by 61% (0.297 to 0.115) and adopts the oracle boundary policy in 89% of worlds.

Engineering a Markov blanket is thus a well-posed problem. It is about locating active-to-sensory paths, choosing whether to cut, model or learn them, and, for an agent engineering its own boundary, preserving the ability to learn what its boundary is.

1 Introduction

A Markov blanket separates internal states μ from external states η through sensory states s and active states a , so that $\mu \perp \eta \mid (s, a)$ [8, 14, 16]. The construct is central to the Free Energy Principle (FEP) and to Active Inference [18]. Its explanatory status is also contested [3, 6]. A recurring worry is that a blanket partition is merely descriptive: it can be read off a system without doing any work in explaining that system’s behaviour.

One natural way to make blankets do work is to treat them as design objects, that is, to ask how behaviour changes when the boundary is engineered differently. As long as the sensory channel is exogenous, however, this reduces to a familiar problem. Comparing two sensory boundaries is then comparing two statistical experiments, which Blackwell’s theorem

[5], Lindley information [17] and Bayesian experimental design [7] already handle. Talk of “blankets” adds vocabulary but no content.

This paper identifies a case where the blanket framing does add content. What distinguishes a Markov blanket from an observation channel is that it contains *active* states, and active states can be causal ancestors of later sensory states. When this happens, the experiment the agent faces is no longer fixed. It depends on what the agent has done, and hence on what the agent has believed. We show that this property is invisible to every criterion defined on per-step channel statistics, yet it changes both the value of the evidence stream and the calibration of any agent that fails to represent it.

Contributions. The paper makes five contributions.

1. *Step-wise equivalence with topological difference* (Proposition 1). We construct a closed-loop boundary and an open-loop boundary that are identical as experiments at every step but differ in whether a sensory state descends from an active state.
2. *Misspecification without detectable marginal error* (Proposition 2). A factorised generative model is exactly correct for the open loop and wrong for the closed loop, although it matches every per-step marginal of both.
3. *Sequential garbling* (Proposition 3). The closed loop is a garbling of its step-equivalent open loop, so the loop reduces decision value even for a perfectly specified agent.
4. *Self-reinforcement threshold* (Proposition 4). An agent that ignores the loop turns it into a positive-feedback system above an analytic threshold κ^* . A correctly specified agent is immune.
5. *Remedies*. We compare structural, inferential and epistemic interventions, including an honest negative result for one-step expected-free-energy probing.
6. *A design map* (Section 7). With explicit costs, we determine when an external designer should leave, cut or learn the loop.
7. *Self-engineering* (Proposition 5 and Section 7.2). An agent that can cut its own loop faces an epistemic trap: cutting destroys the evidence that would justify cutting. We prove that greedy boundary choice can seal itself permanently. We then show that the epistemic term of expected free energy, applied to the agent’s uncertainty about its own blanket topology, avoids the trap.

2 Background and positioning

Exogenous experiments. Blackwell [5] orders experiments by garbling: E' is less informative than E if E' can be generated from E ’s output plus independent noise. Equivalently, E is weakly better in every Bayesian decision problem. Lindley’s measure [17] and Bayesian experimental design [7] quantify and optimise expected information gain. All of these results take the experiment to be fixed before the decision maker acts.

Endogenous and correlated evidence. Several literatures study evidence that is correlated with the recipient’s own past behaviour or beliefs.

- Herding and informational cascades [1, 4]: agents infer from others’ actions, which themselves reflect earlier inferences.

- Persuasion bias [9] and correlation neglect [10]: repeated or correlated signals are treated as independent.
- Misspecified Bayesian learning [2], and in particular Berk–Nash equilibrium [11]: agents with wrong models whose actions shape the data they learn from settle on self-confirming beliefs.
- Sensor fusion: the “data incest” problem of double-counting shared information [15].
- Neuroscience: the refference principle and corollary discharge [20], which address how organisms distinguish self-generated from externally generated sensation.

What is new here. None of these phenomena is new, and we do not claim otherwise. Our contribution is to locate them in a single structural property of a Markov blanket: whether some sensory state has an active state among its causal ancestors. We make that property precise, and we show formally that it escapes per-step channel comparison. We then treat the property as an engineering variable that admits three kinds of remedy: changing the topology, changing the model, or learning the topology.

3 Model

3.1 Agents and states

A binary external state $\eta \in \{0, 1\}$ is drawn uniformly and fixed within an episode of T steps. At each step t the decision agent receives $s_t = (s_{1,t}, s_{2,t})$, updates its belief $q_t(\eta)$, and emits an announcement

$$a_t = \arg \max_{\eta} q_t(\eta), \quad (1)$$

with ties broken uniformly. The announcement is an active state: it is part of the agent’s blanket and is observable by others. Observer A_1 always reports an independent observation of reliability $\rho \in (1/2, 1)$, so that $P(s_{1,t} = \eta \mid \eta) = \rho$.

3.2 Two boundary organisations

Definition 1 (Closed loop, system A). Observer A_2 can see the decision agent’s announcements. At each $t \geq 2$, independently with probability $\kappa \in [0, 1]$, it repeats the previous announcement; otherwise it reports its own independent observation $z_{2,t}$ of reliability ρ :

$$s_{2,t} = C_t a_{t-1} + (1 - C_t) z_{2,t}, \quad C_t \sim \text{Bern}(\kappa). \quad (2)$$

At $t = 1$ there is no previous announcement, and $s_{2,1} = z_{2,1}$.

Let $\alpha_t = P(a_t = \eta \mid \eta)$ denote the accuracy of the announcement process. By the symmetry of the model, α_t does not depend on the value of η .

Definition 2 (Step-matched open loop, system B). Observer A_2 is an independent sensor whose reliability at step t is

$$\rho_t^{\text{eff}} = \kappa \alpha_{t-1} + (1 - \kappa) \rho \quad (t \geq 2), \quad \rho_1^{\text{eff}} = \rho. \quad (3)$$

System B is defined relative to the announcement process of A. It is the open loop that reproduces A’s per-step statistics exactly.

Blanket topology. In A, $s_{2,t} \in \text{de}(a_{t-1})$: a sensory state descends from an active state. The path $s_{<t} \rightarrow \mu_{t-1} \rightarrow a_{t-1} \rightarrow s_{2,t}$ connects past and present sensory states without passing through η . In B no such path exists. The two systems share the same state partition. What differs is the causal topology of the blanket: whether the arrows run from active to sensory states.

3.3 Agents

Definition 3 (Naive agent). The naive agent uses the factorised likelihood

$$p_{\text{naive}}(s_{1:t} \mid \eta) = \prod_{\tau \leq t} p(s_{1,\tau} \mid \eta) \text{Bern}_{\rho_{\tau}^{\text{eff}}}(s_{2,\tau} \mid \eta), \quad (4)$$

with the exact per-step marginal reliability ρ_{τ}^{eff} . In A we compute ρ_{τ}^{eff} as a self-consistent fixed point: the reliability the naive agent's own announcements induce. The naive agent is therefore correct about every per-step marginal it could check, and its model is the best-fitting independent model of the data it generates, in the spirit of [11].

Definition 4 (Aligned agent). The aligned agent represents the active-to-sensory path, as an efferece copy would:

$$p(s_{2,t} \mid \eta, a_{t-1}) = \kappa \mathbb{I}[s_{2,t} = a_{t-1}] + (1 - \kappa) \rho^{\mathbb{I}[s_{2,t} = \eta]} (1 - \rho)^{\mathbb{I}[s_{2,t} \neq \eta]}. \quad (5)$$

4 Theory

Proposition 1 (Step-wise experimental equivalence). *For every t , $p_A(s_t \mid \eta) = p_B(s_t \mid \eta)$. Consequently, no criterion defined on the family of per-step channels $\{p(s_t \mid \eta)\}_{t \leq T}$ distinguishes A from B. This includes the Blackwell ordering of per-step experiments, per-step Lindley information and one-step value of information.*

Proof. The announcement a_{t-1} is a function of $s_{1:t-1}$ and of tie-breaking noise. Given η , these are independent of the current draws $s_{1,t}$, $z_{2,t}$ and C_t . Hence $s_{1,t} \perp s_{2,t} \mid \eta$ in A, as in B. By (2), $P_A(s_{2,t} = \eta \mid \eta) = \kappa \alpha_{t-1} + (1 - \kappa) \rho = \rho_t^{\text{eff}}$, which by (3) equals $P_B(s_{2,t} = \eta \mid \eta)$. The component $s_{1,t}$ has the same law in both systems. $\square$

Proposition 2 (Joint non-equivalence and misspecification). *Let $\kappa > 0$ and $0 < \alpha_{t-1} < 1$. Then $\text{Cov}_A(s_{2,t}, a_{t-1} \mid \eta) = \kappa \text{Var}(a_{t-1} \mid \eta) > 0$, whereas $\text{Cov}_B(s_{2,t}, a_{t-1} \mid \eta) = 0$. Hence $p_A(s_{1:T} \mid \eta) \neq p_B(s_{1:T} \mid \eta)$ for $T \geq 2$. The factorised likelihood of the naive agent is exactly correct in B and misspecified in A.*

Proof. Given η , both C_t and $z_{2,t}$ are independent of a_{t-1} . Therefore $\text{Cov}(C_t a_{t-1} + (1 - C_t) z_{2,t}, a_{t-1} \mid \eta) = \kappa \text{Var}(a_{t-1} \mid \eta)$, which is positive when $0 < \alpha_{t-1} < 1$. In B, $s_{2,t}$ is independent of the past given η . $\square$

Propositions 1 and 2 together say that the difference between A and B is invisible step by step but real over histories. The next result says that it matters even to an agent who models it correctly.

Proposition 3 (The closed loop is a sequential garbling). *Suppose $\alpha_{t-1} \geq \rho$ for all $t \geq 2$. Then the history experiment $s_{1:T}$ of system A is a garbling of the history experiment of system B, where B is matched to A's announcement process. Hence, for every decision problem based on the history and every prior, the Bayes-optimal expected utility under A is at most that under B.*

Proof. We generate A’s history from B’s history and independent noise, step by step. The s_1 components are copied. Given B’s output $s_{2,t}^B$, which has reliability $\rho_t^{\text{eff}} \geq \rho$ because $\alpha_{t-1} \geq \rho$, flip it with probability

$$\phi_t = \frac{\rho_t^{\text{eff}} - \rho}{2\rho_t^{\text{eff}} - 1} \in [0, \frac{1}{2}). \quad (6)$$

This yields an independent reliability- ρ observation $\tilde{z}_{2,t}$. Then, with independent probability κ , replace it by a_{t-1} . Here a_{t-1} is computed by running A’s announcement rule on the already constructed history $\tilde{s}_{1:t-1}$, which is a function of B’s outputs and earlier noise. By induction on t , the constructed sequence has exactly the law of A’s history. Blackwell’s theorem [5] gives the value inequality. $\square$

Remark. Proposition 3 turns the step-wise equivalence of Proposition 1 into a sequential inequality. The loop spends channel capacity re-transmitting the agent’s own past output. This part of the signal carries no information about η beyond what the agent already has. The condition $\alpha_{t-1} \geq \rho$ holds in all our runs.

Proposition 4 (Self-reinforcement threshold). *Consider the naive agent in A with log-odds ℓ_t in favour of its current announcement. Suppose that announcement is wrong, $a_{t-1} \neq \eta$, and let $r = \rho_t^{\text{eff}}$. The expected one-step change of ℓ in favour of the wrong state is*

$$\Delta_{\text{naive}} = \kappa \logit(r) - (2\rho - 1) [\logit(\rho) + (1 - \kappa) \logit(r)], \quad (7)$$

which is positive if and only if

$$\kappa > \kappa^*(\rho, r) = \frac{(2\rho - 1) [\logit(\rho) + \logit(r)]}{2\rho \logit(r)}. \quad (8)$$

For the aligned agent, the expected one-step change of its log-odds in favour of the true state equals a Kullback–Leibler divergence between its likelihoods under η and $-\eta$. This is strictly positive because s_1 is always informative. Its wrong beliefs are therefore never self-reinforcing in expectation.

Proof. The component $s_{1,t}$ contributes $-(2\rho - 1) \logit(\rho)$ in expectation. With probability κ , $s_{2,t} = a_{t-1}$, which the naive agent scores as $+\logit(r)$. With probability $1 - \kappa$, $s_{2,t}$ is an independent reliability- ρ report, which contributes $-(2\rho - 1) \logit(r)$ in expectation. Summing gives (7), and solving $\Delta_{\text{naive}} = 0$ for κ gives (8). For the aligned agent, the expected log-likelihood ratio under the true model is a KL divergence. It is non-negative, and strictly positive because the s_1 term alone contributes $\text{KL}(\text{Bern}_\rho \| \text{Bern}_{1-\rho}) > 0$. $\square$

When the announcement is correct, the loop pushes the naive agent toward the truth as well. Above κ^* , both correct and incorrect beliefs are therefore self-reinforcing: the naive agent becomes a bistable system that locks in on whichever state it favours early.

5 Simulation results

Design. Unless stated otherwise: $\rho = 0.60$, $\kappa = 0.50$, $T = 30$, $N = 200,000$ episodes, seed 20260930. We use common random numbers across conditions. ρ^{eff} is computed as the naive agent’s self-consistent fixed point in A. It rises from 0.600 to 0.681 over the episode.

For system A we report the naive and the aligned agent. Each generates its own announcements and hence its own loop. For system B we report the naive agent, which is exactly specified there. The comparator B is matched to the naive agent’s announcement process. In all runs the aligned agent’s announcement accuracy is at least that of the naive agent, so a comparator matched to the aligned agent would be at least as informative as B. The A-aligned versus B gap reported below is therefore conservative.

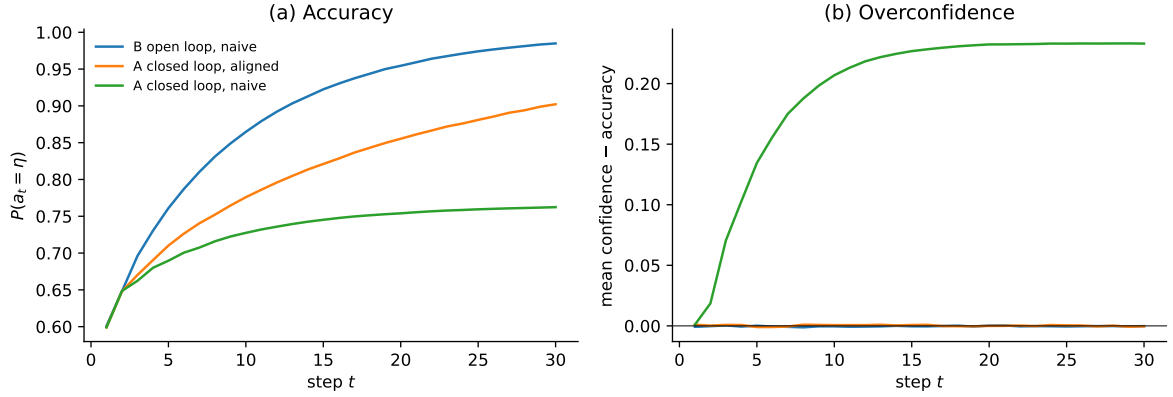


Figure 1: Step-wise equivalent boundaries with different topology. (a) Accuracy over time. (b) Mean confidence minus accuracy. Only the naive agent in the closed loop is miscalibrated.

5.1 Equivalence, value loss and miscalibration

The largest per-step difference between the empirical marginals $P(s_{2,t} = \eta)$ in A and B is 0.0021. This is Monte Carlo noise, as Proposition 1 requires.

Condition	Accuracy	Mean confidence	Lock-in	Log-loss
B, open loop, naive (exact)	0.985	0.985	0.0006	0.041
A, closed loop, aligned	0.902	0.902	0.0010	0.237
A, closed loop, naive	0.762	0.995	0.193	1.838

Table 1: Final step ($T = 30$). Lock-in is the fraction of episodes ending with $q(\eta_{\text{true}}) < 0.01$. A and B are identical experiments at every step.

Table 1 and Figure 1 give three results.

Result 1: topology reduces value, even with a correct model. The aligned agent in A reaches 90.2% accuracy against 98.5% in B, with identical per-step channels. This is Proposition 3 made quantitative.

Result 2: ignoring the topology produces confident error. The naive agent in A reports a mean confidence of 0.995 while being right only 76.2% of the time. It ends with more than 99% confidence in the wrong state in 19.3% of episodes. The same model is perfectly calibrated in B.

Result 3: early errors become permanent. In A the naive agent is wrong at step 10 in 27.3% of episodes. Of those, 85.6% are still wrong at step 30. For the aligned agent the corresponding persistence is 26.7%. In B it is 6.5%.

5.2 Dependence on κ and robustness

At $\kappa = 0$ all conditions coincide (accuracy 0.940–0.941), as they must; Figure 2 shows the full sweep. As κ increases from 0.1 to 0.9:

- naive accuracy in A falls from 0.926 to 0.627, and lock-in rises from 0.007 to 0.371;
- the aligned agent degrades gradually, from 0.932 to 0.875, with lock-in never above 0.0013;
- the matched open loop B stays between 0.961 and 0.986.

The same pattern holds across sensory reliability at $\kappa = 0.5$ (Table 2).

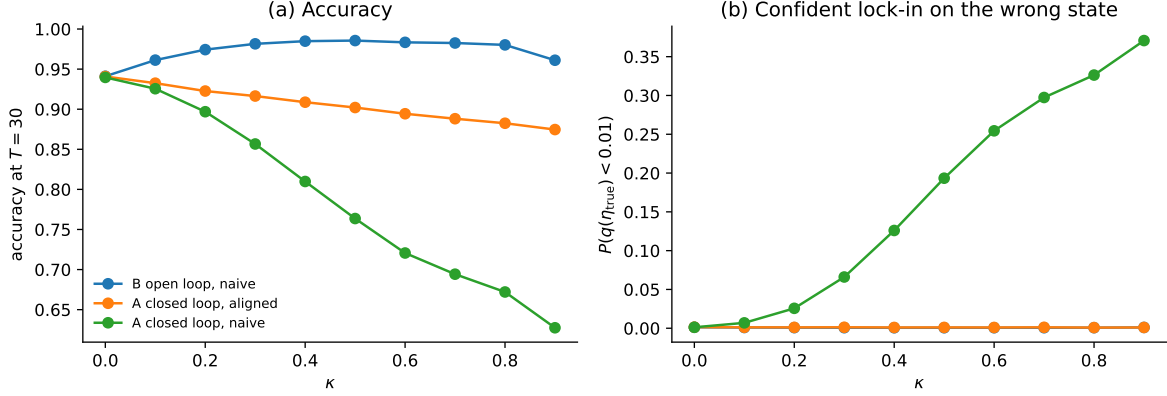


Figure 2: Sweep over the loop strength κ ($N = 100,000$ per point). At $\kappa = 0$ all conditions coincide.

	$\rho = 0.55$	$\rho = 0.60$	$\rho = 0.70$
B, naive (exact)	0.859	0.986	1.000
A, aligned	0.737	0.902	0.996
A, naive	0.637	0.764	0.936
A, naive: lock-in	0.175	0.193	0.054

Table 2: Final accuracy at $\kappa = 0.5$ for three sensory reliabilities ($N = 100,000$).

5.3 The self-reinforcement threshold

We evaluate Eq. (8) at the late-episode value of ρ^{eff} for each κ . This gives $\kappa^* \in [0.25, 0.30]$. The analytic drift changes sign between $\kappa = 0.2$ ($\Delta = -0.054$ nats per step) and $\kappa = 0.3$ ($\Delta = +0.039$) (Figure 3). Simulated lock-in accelerates in exactly this interval: 0.026, then 0.066, then 0.126 at $\kappa = 0.2, 0.3, 0.4$. Below the threshold, lock-in is small but non-zero, because of finite-horizon fluctuations. The threshold marks where confident error stops being a transient and becomes an attractor.

6 Three remedies

The same defect admits interventions of three different kinds.

Structural. Change the blanket topology by cutting $a \rightarrow z_2$, so that $\kappa = 0$.

Inferential. Keep the topology and align the generative model, using Eq. (5) with the known κ .

Epistemic. Keep the topology and let the agent learn it. The agent holds a joint belief over (η, κ) on the grid $\kappa \in \{0, 0.1, \dots, 0.9\}$, with a uniform prior.

We test three variants of the epistemic learner.

- *Passive*: always announce the most probable state.
- *Random probes*: announce a uniformly random state with probability ε .
- *EFE probes*: choose the announcement x by minimising a one-step reduced expected free energy

$$G(x) = \underbrace{1 - q(\eta = x)}_{\text{risk}} - \beta \underbrace{\mathbb{E}_{q(o|x)} [D_{\text{KL}}(q(\eta, \kappa | o, x) \| q(\eta, \kappa))]}_{\text{information gain on states and parameters}}. \quad (9)$$

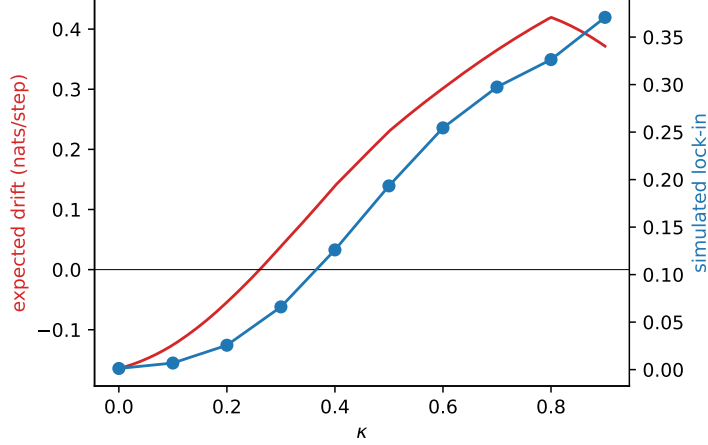


Figure 3: Analytic expected drift toward the wrong state for the naive agent (Eq. 7, left axis) and simulated lock-in (right axis).

This contains the parameter-information (“novelty”) term of Active Inference [13, 19].

Announcing against one’s current belief is informative about κ : an echo of a contrarian announcement is distinguishable from independent evidence. Probing therefore trades announcement accuracy for structural information.

Remedy	Final acc.	Ann. acc.	Probe rate	$ \hat{\kappa} - \kappa $
None (naive)	0.762	—	—	—
Structural (cut loop)	0.940	—	—	—
Inferential (known κ)	0.902	—	—	—
Epistemic, passive	0.890	0.798	0	0.127
Epistemic, random $\varepsilon = 0.1$	0.891	0.771	0.050	0.126
Epistemic, random $\varepsilon = 0.3$	0.897	0.712	0.150	0.123
Epistemic, EFE $\beta = 1$	0.892	0.799	0.007	0.127
Epistemic, EFE $\beta = 4$	0.891	0.800	0.013	0.127

Table 3: Remedies at $\rho = 0.6$, $\kappa = 0.5$, $T = 30$ ($N = 50,000$). “Ann. acc.” is the mean per-step accuracy of announcements, which is the pragmatic cost of probing.

Table 3 and Figure 4 give three results.

Result 4: the structural remedy dominates in accuracy. Cutting the loop (94.0%) beats modelling it (90.2%). This follows from Proposition 3: modelling the loop lets the agent discount echoes, but only cutting it restores the fresh evidence those echoes displaced. Whether cutting is optimal therefore depends on its cost relative to modelling. This is the engineering trade-off.

Result 5: learning the topology passively recovers most of the loss. Holding uncertainty over κ and marginalising, without any probing, already gives 89.0%. That is 91% of the gap between the naive agent (76.2%) and the agent that knows κ (90.2%). The same holds at $\kappa = 0.2$ (0.920 against 0.923 known and 0.898 naive) and at $\kappa = 0.8$ (0.875 against 0.883 known and 0.673 naive). The crucial step is representing the possibility of a loop, not knowing its strength.

Result 6: one-step EFE probing is almost never worth it here. The EFE agent probes in under 1.5% of steps even at $\beta = 4$, and its accuracy is indistinguishable from the passive learner’s. Random probing at $\varepsilon = 0.3$ improves final accuracy by 0.7 points, but it costs 8.6 points of announcement accuracy, and the posterior over κ remains broad (entropy ≈ 1.83

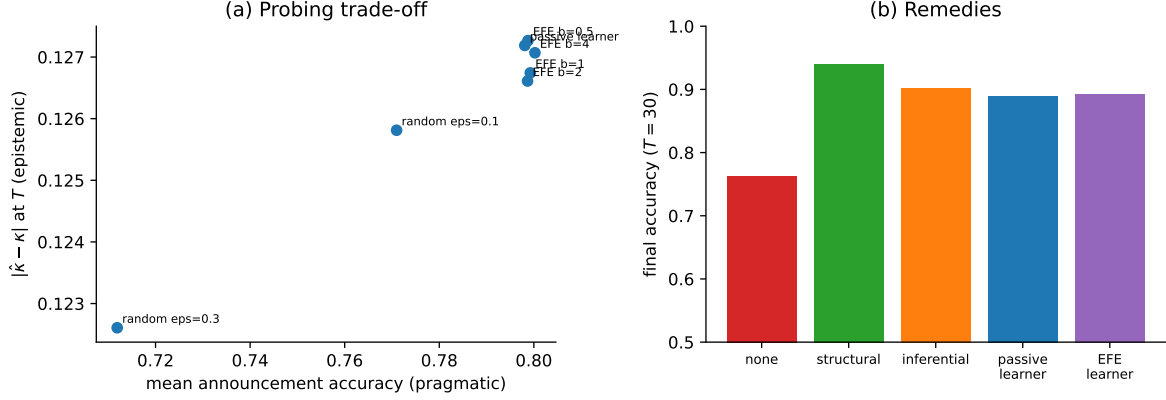


Figure 4: (a) Pragmatic cost versus structural information for probing strategies. (b) Final accuracy under the three remedies.

nats against a maximum of $\ln 10 \approx 2.30$). By its own criterion, the EFE agent is right not to probe. With one-step lookahead, the information about κ gained from a single contrarian announcement is small compared with its immediate risk.

We read Result 6 as a genuine negative result about myopic epistemic action, not as a failure of the framework. The benefit of knowing κ accrues over many future steps, and a one-step objective cannot see it. Multi-step policies, or longer interactions in which κ persists across episodes, are the natural setting in which active probing of one’s own blanket should pay off. We leave this to future work.

7 Engineering the blanket

The previous section compared remedies by accuracy alone. Engineering requires costs. We now treat the blanket’s topology as the design variable, first from the perspective of an external designer, and then from the perspective of the agent itself.

7.1 A design map

An external designer chooses one of three configurations for a system with loop strength κ :

- *none*: leave the loop and a naive agent, at no cost;
- *cut*: remove the path $a \rightarrow z_2$, at cost C_{cut} ;
- *learn*: keep the loop and equip the agent with the joint (η, κ) model, at cost C_{model} (computation, memory, model complexity).

The design objective is

$$J(d; \kappa) = [1 - \text{Acc}_d(\kappa)] + C_d, \quad d^*(\kappa) = \arg \min_d J(d; \kappa), \quad (10)$$

where the accuracies are the simulated final accuracies at $T = 30$.

Figure 5 shows three regimes.

- When $\kappa \approx 0$, doing nothing is optimal: there is no loop to remedy.
- When cutting is cheap relative to the damage the loop causes, cutting wins. Its region grows with κ , because the value lost to the loop grows with κ (Proposition 3).

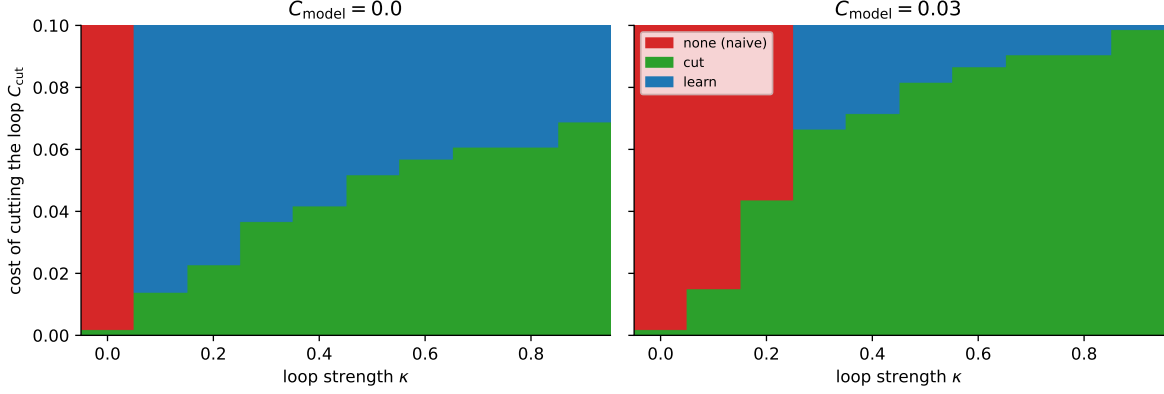


Figure 5: Design map. The optimal blanket configuration as a function of loop strength κ and the cost of cutting the loop, for two costs of the richer generative model.

- When cutting is expensive, learning wins, unless the model cost exceeds what the loop destroys. With $C_{\text{model}} = 0.03$, leaving a weak loop ($\kappa \leq 0.2$) unmodelled becomes optimal.

This is the analogue, for blanket topology, of the communication-cost trade-off in classical sensor design. Here, though, the benefit side is governed by the sequential results of Section 4, not by per-step channel comparison.

7.2 Self-engineering: the agent chooses its own boundary

We now remove the external designer. The agent lives through a sequence of $M = 40$ episodes in a world whose loop strength κ is fixed but unknown. η is redrawn in each episode. Before each episode the agent chooses a boundary configuration:

- *visible*: its announcements are observable, so the loop, of unknown strength, is active;
- *hidden*: it withholds its announcements, which cuts the loop, at a cost C per episode.

Within an episode the agent is the joint (η, κ) learner of Section 6. Its belief $q(\kappa)$ is carried across episodes.

Policy selection. The agent evaluates the two boundary policies with

$$G(\text{visible}) = \mathbb{E}_{q(\kappa)} [1 - \text{Acc}_{\text{learn}}(\kappa)] - \beta \mathcal{I}_{\kappa}(\text{visible}), \quad (11)$$

$$G(\text{hidden}) = [1 - \text{Acc}_{\text{hidden}}] + C, \quad (12)$$

and chooses the minimum. Here $\mathcal{I}_{\kappa}(\text{visible})$ is the expected information gain about κ from one visible episode. It is estimated by Monte Carlo from a bank of simulated episodes: κ is sampled from q , a simulated episode is drawn, and q is updated with that episode's likelihood over κ . The case $\beta = 0$ is greedy expected utility. For $\beta > 0$ the objective contains the epistemic, parameter-information term of expected free energy [13, 19], applied to the agent's uncertainty about the topology of its own blanket.

Proposition 5 (Self-sealing boundaries). *Under the hidden configuration, the likelihood of every episode is constant in κ , so $q(\kappa)$ is unchanged. Consequently, if a greedy agent ($\beta = 0$) chooses hidden at some episode, it chooses hidden at every later episode, whatever the true κ .*

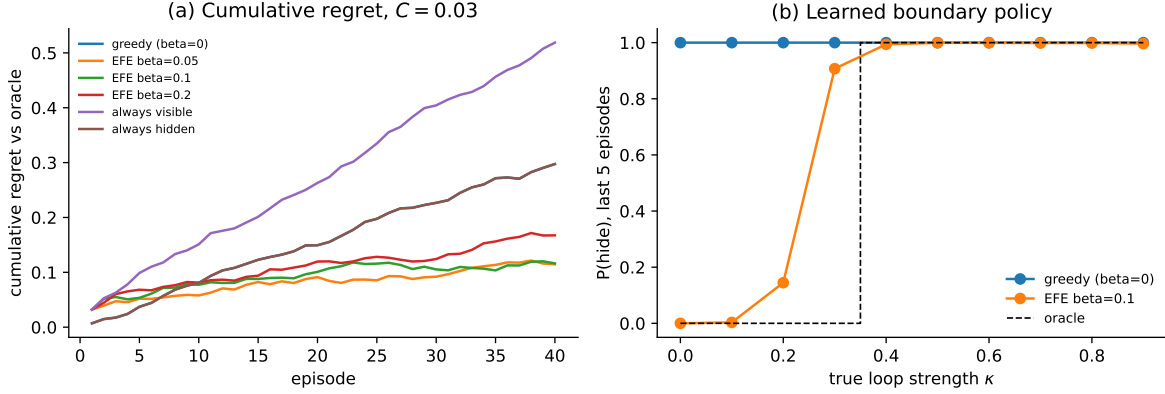


Figure 6: Self-engineering at $C = 0.03$. (a) Cumulative regret. The greedy agent coincides with “always hidden”. (b) Probability of hiding over the last five episodes as a function of the true κ , against the oracle policy.

Proof. When announcements are withheld, $s_{2,t}$ is an independent reliability- ρ report for every value of κ . The episode likelihood therefore factorises into a term in η and a term constant in κ , and the posterior over κ equals the prior. With $\beta = 0$, both (11) and (12) depend on the history only through $q(\kappa)$, which no longer changes. The inequality that selected *hidden* is therefore reproduced at every later episode. $\square$

Proposition 5 is the engineering counterpart of Proposition 4. There, an agent that ignores its loop locks into a wrong belief about the world. Here, an agent that cuts its loop too early locks into an unrevisable belief about itself. Cutting is the best remedy for the loop’s informational damage (Result 4), but it is epistemically blind: it removes the only evidence about whether it was needed. For $\beta > 0$, $\mathcal{I}_\kappa(\text{visible}) > 0$ whenever $q(\kappa)$ is not degenerate. This lowers $G(\text{visible})$, so the agent keeps its boundary open long enough to learn what it is.

Setup. We simulate 3,000 worlds, with 300 for each $\kappa \in \{0, 0.1, \dots, 0.9\}$. The prior over κ is uniform, and $M = 40$ episodes of $T = 30$ steps are run for each world. Regret is measured against an oracle that knows κ and chooses the better of the known- κ agent and the hidden configuration. Utility per episode is final correctness minus C if hidden.

Policy	$C = 0.01$		$C = 0.03$		$C = 0.05$	
	Regret	Match	Regret	Match	Regret	Match
Always visible	1.078	0.20	0.519	0.40	0.180	0.70
Always hidden	0.057	0.80	0.297	0.60	0.759	0.30
Greedy ($\beta = 0$)	0.057	0.80	0.297	0.60	0.152	0.76
EFE, $\beta = 0.05$	0.080	0.85	0.115	0.87	0.163	0.81
EFE, $\beta = 0.1$	0.060	0.89	0.116	0.89	0.115	0.83
EFE, $\beta = 0.2$	0.180	0.92	0.167	0.91	0.144	0.85

Table 4: Self-engineering over 40 episodes. Regret is cumulative, relative to the oracle. Match is the fraction of worlds whose policy over the last five episodes coincides with the oracle’s.

Table 4 and Figure 6 give three results.

Result 7: greedy self-engineering seals itself. At $C = 0.03$, the greedy agent’s prior expected gain from hiding exceeds the cost. It therefore hides in every world from the first episode and never stops, exactly as Proposition 5 predicts. Its posterior entropy over κ stays at its maximum

($\ln 10 \approx 2.30$). It pays for a cut it does not need in the 40% of worlds where $\kappa \leq 0.3$, and it is indistinguishable from “always hidden” (regret 0.297).

Result 8: the epistemic term breaks the seal. With $\beta = 0.05$ or 0.1 , the agent starts visible in every world, learns κ , and then hides selectively. Regret falls from 0.297 to 0.115, a 61% reduction. The final boundary policy matches the oracle in 89% of worlds (Figure 6b). Worlds with $\kappa \leq 0.1$ stay open, worlds with $\kappa \geq 0.4$ close, and disagreements concentrate at the boundary $\kappa \in \{0.2, 0.3\}$. There, the agent’s own forecast, which is based on the passive learner’s accuracy, is slightly more pessimistic about the open loop than the oracle is. At $C = 0.05$ the greedy agent happens to start visible and learns passively. Even so, the epistemic term reduces regret by 24% (0.152 to 0.115).

Result 9: exploration has a price. When cutting is cheap ($C = 0.01$), hiding is optimal in 80% of worlds, and greedy sealing is close to optimal (regret 0.057). A moderate epistemic weight matches it ($\beta = 0.1$: 0.060). A large weight over-explores and triples regret ($\beta = 0.2$: 0.180). The epistemic term is valuable exactly when an irreversible-looking structural choice would otherwise be made under uncertainty, and not by default.

Reconciling Results 6 and 8. One-step expected-free-energy probing within an episode was almost never worth it (Result 6). Epistemic value at the level of boundary policies across episodes is decisive (Result 8). The difference is the timescale at which the information is used. A single contrarian announcement yields little information about κ relative to its immediate cost. A whole visible episode yields enough to decide a configuration that will be kept for many episodes. Epistemic action about one’s own blanket pays off at the timescale at which the blanket itself is reconfigured.

8 Discussion

What the blanket framing contributes. Propositions 1–3 identify a property of a boundary that matters for inference and value but cannot be recovered from its per-step channel statistics. That property is the presence of active states among the causal ancestors of sensory states. It is precisely the feature that separates a Markov blanket from an exogenous observation channel. In this setting the blanket is not a relabelling of the channel. It is the object whose topology determines whether per-step channel analysis is valid at all.

A design principle. The results suggest a simple principle for engineering blankets:

For every sensory state, determine whether it has active-state ancestors. Where it does, either cut the path, model it, or learn it. If the system chooses for itself, do not let it cut what it has not yet learned.

The three options differ in cost and effectiveness. Cutting restores information. Modelling restores calibration but not information. Learning restores most of the calibration without prior knowledge of the loop strength. The design map (Figure 5) makes the choice among them a cost-benefit calculation. The last clause of the principle is Proposition 5: structural remedies are epistemically blind, so a self-engineering system must value information about its own boundary before closing it.

Relation to Active Inference. In Active Inference, beliefs about the consequences of one’s own actions are part of the generative model [12, 18]. Our results make one such belief explicit: how much of what I sense is my own echo. The aligned model (5) is a minimal efference copy [20]. The joint (η, κ) learner is a minimal form of structure learning. The self-engineering agent of Section 7.2 goes one step further. It selects policies over its own blanket topology, and the

epistemic term of expected free energy, applied to uncertainty about that topology, is what prevents it from sealing its boundary on unrevisable beliefs. The contrast between Results 6 and 8 locates where epistemic value about one’s own blanket matters: at the timescale at which the blanket is reconfigured, not at the timescale of individual actions. This connects the formal results to a central theme of the FEP literature, namely that living systems do not merely have blankets but maintain and remodel them [16]. Our analysis adds a specific constraint on how that remodelling must be done if it is to remain adaptive.

Relation to economics and sensor fusion. The naive agent’s dynamics are those of correlation neglect [10] and persuasion bias [9] applied to one’s own output. Its fixed-point channel model is a self-confirming misspecification in the sense of [11]. The structural remedy corresponds to eliminating data incest [15] at its source, rather than compensating for it downstream. What the present formulation adds is the step-wise equivalence result. It explains why such defects are invisible to any analysis that compares observation channels one step at a time.

Limitations.

- The model is minimal: binary state, stationary environment, known ρ , a single loop, greedy announcements and $T = 30$.
- The comparator B is defined relative to a given announcement process, so it is a construction rather than a physical alternative.
- Proposition 3 requires $\alpha_{t-1} \geq \rho$, which holds in our runs but not universally.
- The within-episode EFE agent uses a one-step horizon and a reduced functional.
- In the self-engineering experiment, the agent forecasts the value of the open loop from the accuracy of the uniform-prior learner, and estimates information gain from a bank of simulated episodes. Both are approximations to full planning. The design map uses simulated accuracies, not closed-form ones.
- Costs are expressed in accuracy units and are stipulated, not derived from a physical model.
- The analysis concerns one kind of topological change. Other active-to-sensory structures, such as loops through the environment rather than through another agent, remain to be studied.

9 Conclusion

Two boundaries can be the same experiment at every step and still differ in a way that matters. When sensory states descend from the agent’s own active states, the boundary is less informative over time than its step-wise statistics suggest, even for a perfectly specified agent. An agent that does not represent this is driven, above a computable threshold, into confident and persistent error.

This is a case in which the Markov blanket, understood as a causal topology including active states, does work that channel-based information theory cannot do. It also turns “engineering Markov blankets” into a concrete design problem at two levels. For an external designer, the problem is to locate active-to-sensory paths and to decide, at given costs, whether to cut, model or learn them. For a system that engineers its own blanket, there is an additional requirement. Cutting a loop destroys the evidence about the loop. A greedy self-engineer can therefore seal its boundary permanently on beliefs it can no longer revise. Valuing information about one’s

own boundary, as the epistemic term of expected free energy does, avoids this trap. In our experiments it reduces regret by up to 61%.

A Reproducibility

All results are produced by the Python package `emb`, which accompanies this paper (module `emb.experiments`; command `python -m emb.experiments`). It uses Python, NumPy and Matplotlib, seed 20260930, and no external data. Sections 5–6 correspond to `run_v8` and Section 7 to `run_v9`. All reported numbers are written to `results_v8.json` and `results_v9.json`.

References

- [1] A. V. Banerjee, “A simple model of herd behavior,” *Quarterly Journal of Economics*, 107(3):797–817, 1992.
- [2] R. H. Berk, “Limiting behavior of posterior distributions when the model is incorrect,” *Annals of Mathematical Statistics*, 37(1):51–58, 1966.
- [3] M. Biehl, F. A. Pollock, and R. Kanai, “A technical critique of some parts of the free energy principle,” *Entropy*, 23(3):293, 2021.
- [4] S. Bikhchandani, D. Hirshleifer, and I. Welch, “A theory of fads, fashion, custom, and cultural change as informational cascades,” *Journal of Political Economy*, 100(5):992–1026, 1992.
- [5] D. Blackwell, “Equivalent comparisons of experiments,” *Annals of Mathematical Statistics*, 24(2):265–272, 1953.
- [6] J. Bruineberg, K. Dołęga, J. Dewhurst, and M. Baltieri, “The emperor’s new Markov blankets,” *Behavioral and Brain Sciences*, 45:e183, 2022.
- [7] K. Chaloner and I. Verdinelli, “Bayesian experimental design: A review,” *Statistical Science*, 10(3):273–304, 1995.
- [8] L. Da Costa, K. Friston, C. Heins, and G. A. Pavliotis, “Bayesian mechanics for stationary processes,” *Proceedings of the Royal Society A*, 477(2256):20210518, 2021.
- [9] P. M. DeMarzo, D. Vayanos, and J. Zwiebel, “Persuasion bias, social influence, and unidimensional opinions,” *Quarterly Journal of Economics*, 118(3):909–968, 2003.
- [10] B. Enke and F. Zimmermann, “Correlation neglect in belief formation,” *Review of Economic Studies*, 86(1):313–332, 2019.
- [11] I. Esponda and D. Pouzo, “Berk–Nash equilibrium: A framework for modeling agents with misspecified models,” *Econometrica*, 84(3):1093–1130, 2016.
- [12] K. Friston, F. Rigoli, D. Ognibene, C. Mathys, T. Fitzgerald, and G. Pezzulo, “Active inference and epistemic value,” *Cognitive Neuroscience*, 6(4):187–214, 2015.
- [13] K. J. Friston, M. Lin, C. D. Frith, G. Pezzulo, J. A. Hobson, and S. Ondobaka, “Active inference, curiosity and insight,” *Neural Computation*, 29(10):2633–2683, 2017.
- [14] K. Friston, L. Da Costa, N. Sajid, C. Heins, K. Ueltzhöffer, G. A. Pavliotis, and T. Parr, “The free energy principle made simpler but not too simple,” *Physics Reports*, 1024:1–29, 2023.
- [15] S. J. Julier and J. K. Uhlmann, “A non-divergent estimation algorithm in the presence of unknown correlations,” in *Proceedings of the American Control Conference*, vol. 4, pp. 2369–2373, 1997.
- [16] M. Kirchhoff, T. Parr, E. Palacios, K. Friston, and J. Kiverstein, “The Markov blankets of life: autonomy, active inference and the free energy principle,” *Journal of the Royal Society Interface*, 15(138):20170792, 2018.

- [17] D. V. Lindley, “On a measure of the information provided by an experiment,” *Annals of Mathematical Statistics*, 27(4):986–1005, 1956.
- [18] T. Parr, G. Pezzulo, and K. J. Friston, *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.
- [19] P. Schwartenbeck, J. Passetker, T. U. Hauser, T. H. B. FitzGerald, M. Kronbichler, and K. J. Friston, “Computational mechanisms of curiosity and goal-directed exploration,” *eLife*, 8:e41703, 2019.
- [20] E. von Holst and H. Mittelstaedt, “Das Reafferenzprinzip,” *Naturwissenschaften*, 37:464–476, 1950.